DATA ENGINEER — 6 MONTH INTERNSHIP ROADMAP

4 Hrs/Day | 720 Hours | Mini Project Per Topic | Capstone: CP Intelligence Platform

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STRUCTURE

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Every topic          → comes with a mini project (build same day)
Every 3-4 weeks      → a checkpoint project (combines the topics)
Final month          → the CP Contest Intelligence Platform (capstone)
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MONTH 1 — PYTHON + SQL + LINUX BASICS

Week 1 — Python Core (28 hrs)

Day	Topic	Mini Project
1	Variables, data types, f-strings	Unit Converter (km↔miles, °C↔°F)
2	if/elif/else, operators	Grade Calculator
3	for/while loops	Number Pattern Printer
4	Functions, return, default args	Simple Calculator (4 functions)
5	*args, **kwargs	Flexible Average Calculator
6	Lists, tuples	To-Do List CLI
7	Sets, dicts	Word Frequency Counter

Checkpoint: Menu-driven CLI combining all 7 tools.

Week 2 — Python Applied (28 hrs)

Day	Topic	Mini Project
8	File I/O (text, CSV)	Daily Journal App
9	JSON read/write	Contact Book
10	try/except/finally	Safe Divider (never crashes)

Day	Topic	Mini Project
11	OOP — classes, **init**	Bank Account Simulator
12	OOP — inheritance	Vehicle Hierarchy (Car/Bike)
13	List comprehension, lambda, map, filter	Data Cleaner (one-liners)
14	Logging module	Logged Calculator

🎯 **PROJECT 1: "Personal Finance Tracker CLI"** reader.py → categorizer.py → calculator.py → reporter.py → main.py

Week 3 — SQL Foundations (28 hrs)

Day	Topic	Mini Project
15	SELECT, WHERE, ORDER BY, LIMIT, DISTINCT	Movie DB — 10 filter queries
16	All JOINs	Student-Course Enrollment DB
17	Aggregations, GROUP BY, HAVING	Sales Report Generator
18	Subqueries (scalar, correlated, EXISTS)	Above Average Salary Finder
19	Window Functions (ROW_NUMBER, RANK, LAG, LEAD)	Leaderboard with rank change
20	Running totals, PARTITION BY	Running Bank Balance
21	CTEs (single, chained, recursive)	Employee Hierarchy Explorer

Week 4 — PostgreSQL Deep (28 hrs)

Day	Topic	Mini Project
22	Setup, data types, JSONB	Product Catalog with JSONB
23	DDL, constraints	Library Management Schema
24	Indexes, EXPLAIN ANALYZE	Query Speed Test (100K rows)
25	Transactions, ACID	Money Transfer Simulator
26	Normalization, Star Schema	E-commerce Schema Design
27	Query optimization	Slow Query Fixer (5 queries)
28	LeetCode Database practice	15 problems (Easy + Medium)

🎯 **PROJECT 2: "SQL Murder Mystery + Custom Dataset"** Solve the mystery + build your own 500-row dataset + 20 complex queries.

Linux (spread evenings, 8 hrs): File Organizer Script, Log Analyzer, Background Task Runner.

MONTH 2 — PYTHON FOR DE + DATA MODELING + ETL

Week 5 — pandas (28 hrs)

Day	Topic	Mini Project
29-30	pandas basics	CSV Explorer Tool
31-32	GroupBy, merge, pivot_table	Sales Pivot Reporter
33-34	NaN handling, datetime, strings	Messy Data Cleaner
35	pyarrow + Parquet	CSV vs Parquet Benchmark

Week 6 — APIs + DB Connections (28 hrs)

Day	Topic	Mini Project
36	requests — APIs, pagination	Weather Fetcher (10 cities)
37	pydantic validation	API Response Validator
38	psycopg2 + SQLAlchemy	JSON to PostgreSQL Loader
39	pytest basics	Test Suite for Loader
40	Project structure, .env, logging	Refactor Weather Tool
41-42	Modular ETL build	Weather ETL Pipeline v1

Week 7 — Data Modeling (28 hrs)

Day	Topic	Mini Project
43-44	OLTP vs OLAP, Star/Snowflake Schema	Design 3 schemas (gym, food, library)
45-46	ETL vs ELT	Build same pipeline both ways

Day	Topic	Mini Project
47	Idempotency, upsert logic	Idempotent Loader
48	Data Quality (5 checks)	Reusable DataQualityChecker class

Week 8 — ETL Advanced (28 hrs)

Day	Topic	Mini Project
49	Batch vs Streaming	1-page architecture comparison doc
50-51	SCD Type 1/2/3	Price History Tracker (SCD Type 2)
52-54	Lineage + Monitoring	pipeline_runs monitoring table
55-56	Late data + Backfill	Backfill script (30 missing days)

🎯 **PROJECT 3: "Stock Price ETL Pipeline"** Yahoo Finance API → pydantic → star schema → PostgreSQL → 10 queries → monitoring.

MONTH 3 — DOCKER + AIRFLOW

Week 9-10 — Docker (28 hrs)

Day	Topic	Mini Project
57-58	Containers, images, Dockerfile	Dockerize Hello World
59-60	docker-compose, volumes, networks	PostgreSQL in Docker
61-62	Write your own Dockerfile	Dockerize Stock Pipeline
63	Full stack compose	Two-Container App

Week 11 — Airflow Basics (28 hrs)

Day	Topic	Mini Project
64-65	Architecture, UI tour	First Airflow Setup
66-67	DAG structure, PythonOperator	3-Task Hello DAG
68-69	Sensors, PostgresOperator	Wait-For-File DAG

Week 12 — Scheduling + Errors (14 hrs)

Day	Topic	Mini Project
70-71	Scheduling, cron, backfill	Hourly Weather DAG
72-73	Retries, callbacks	Deliberately Broken DAG (3 retries)

Week 13 — Airflow Advanced (14 hrs)

Day	Topic	Mini Project
74-75	XComs, Variables, Connections	Cross-Task Data Passing DAG
76-77	Dynamic DAGs, TaskGroups	Multi-City Weather DAG (10 cities)

Week 14 — Full Project (28 hrs)

🎯 **PROJECT 4: "Automated News Sentiment Pipeline"** HttpSensor → fetch → sentiment → quality check → load → report → monitor. Runs hourly, idempotent, retry logic, full monitoring.

MONTH 4 — GCP + AWS

Week 15 — GCP Setup + Storage (14 hrs)

Day	Topic	Mini Project
85-86	GCP account, gcloud CLI	GCP Hello World
87-88	Cloud Storage, Python client	GCS Upload/Download Tool

Week 16 — BigQuery (21 hrs)

Day	Topic	Mini Project
89-91	Datasets, SQL, partitioning, clustering	Stock Data in BigQuery (partitioned)
92-93	Nested fields, BQ ML	BigQuery ML — simple regression

Week 17 — IAM + Serverless (21 hrs)

Day	Topic	Mini Project
94-95	IAM, service accounts	Secure Service Account Setup
96-97	Cloud Functions	Auto-Trigger Function (on GCS upload)
98-99	Pub/Sub	Pub/Sub Notifier

Week 18 — Dashboard (14 hrs)

Day	Topic	Mini Project
100-101	Looker Studio	Stock Dashboard v1 (4 charts)

Week 19-20 — AWS Basics (36 hrs)

Day	Topic	Mini Project
102-103	AWS setup, IAM, CLI	AWS Hello World
104-105	S3 + boto3	S3 Mirror Tool
106-107	Redshift basics	Redshift Query Test
108-109	AWS Glue basics	Glue Crawler Demo
110-112	AWS Lambda	S3-Triggered Lambda

🎯 PROJECT 5 (MAIN): "AI Token Cost Analytics Pipeline"

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Proxy Logger (wraps LLM API calls)
→ GCS raw/clean layers
→ Airflow hourly DAG
→ BigQuery (partitioned + clustered)
→ SCD Type 2 for model pricing changes
→ Looker Studio dashboard
→ Slack cost alerts
→ GitHub Actions CI
→ Deployed on GCP
```

MONTH 5 — DBT + SPARK + KAFKA

Week 21 — dbt Basics (21 hrs)

Day	Topic	Mini Project
113-114	What is dbt, ELT philosophy	First dbt Project (default model)
115-116	Project structure	Staging models for stock data
117-118	Staging→Intermediate→Mart	Full dbt model chain

Week 22 — dbt Advanced (21 hrs)

Day	Topic	Mini Project
119-120	dbt tests	10 tests across models
121-122	Incremental models, macros	Incremental daily load model
123-124	Airflow + dbt	🎯 **PROJECT 6: dbt Layer for Token Analytics**

Week 23-24 — Spark (56 hrs)

Day	Topic	Mini Project
125-126	Architecture, PySpark intro	PySpark Hello World
127-128	DataFrame operations	Big Stock Data Processor (5M rows)
129-130	PySpark SQL, window functions	PySpark Window Functions
131-132	Performance tuning	Optimize the Big Job
133-134	Spark on GCP (Dataproc)	Dataproc Job Submission

Week 25 — Kafka (20 hrs)

Day	Topic	Mini Project
135-136	Concepts — topics, producers, consumers	Kafka Hello World
137-138	Python producer + consumer	Live Event Simulator

Day	Topic	Mini Project
139	Kafka + DE use cases	Cost Tracker Stream (LLM events)

MONTH 6 — DSA + SQL INTERVIEWS + CAPSTONE

Week 26-27 — DSA (56 hrs)

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Arrays (Two Pointer, Sliding Window, Prefix Sum)  → 8 problems
Strings (palindrome, anagram)                     → 6 problems
HashMaps + HashSets                               → 6 problems
Sorting (implement Merge + Quick Sort once)
Linked Lists                                      → 6 problems
Trees (BFS, DFS, BST)                             → 6 problems
Stacks + Queues                                   → 5 problems
Recursion                                         → 4 problems
```

Target: 60-80 LeetCode problems total (40 Easy, 30 Medium, 10 Hard)

Week 28 — SQL Interview Prep (28 hrs)

```
Window function problems     → 10 problems
Complex JOIN problems        → 10 problems
Aggregation problems         → 10 problems
LeetCode DB + DataLemur + stratascratch → mixed hard problems
```

Week 29 — System Design for DE (28 hrs)

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5-step design framework practiced on paper
4 common design questions answered out loud (recorded)
5 "why" questions answered without notes:
  - Why Airflow over cron?
  - Why Parquet over CSV?
  - Why BigQuery over PostgreSQL?
  - What is idempotency and why does it matter?
  - How do you handle a pipeline failing at step 3?
Lambda vs Kappa architecture — 1 page note
```

Week 30 — CAPSTONE + APPLY

🏆 FINAL PROJECT: "CP Contest Intelligence Platform"

Day	Build
168	Ingestion — Codeforces API → pydantic → GCS
169	BigQuery schema — problems, contests, submissions, users (partitioned)
170	dbt models — staging → intermediate → marts
171	Airflow DAG — daily ingestion + weekly retrain
172	ML layer — difficulty predictor + recommender (scikit-learn + MLflow)
173	FastAPI `/recommend` endpoint + Looker Studio dashboard
174-175	GitHub Actions CI, README, architecture diagram, demo video, deploy to Cloud Run

Apply

```
176     → Polish GitHub, LinkedIn, resume
177-178 → Cold outreach — DM 50 data engineers/CTOs on LinkedIn
179     → Mock interviews — 2 SQL, 2 DSA, 1 system design (recorded)
180     → Apply to 50+ companies
```

TOOLS INSTALL ORDER

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Month 1 → Python 3.11, VS Code, PostgreSQL, pgAdmin, Git
Month 2 → pandas, pyarrow, requests, pydantic, psycopg2
Month 3 → Docker Desktop, Docker Compose
Month 4 → gcloud CLI, AWS CLI, BigQuery Python client
Month 5 → dbt-core, dbt-bigquery, PySpark, kafka-python
Month 6 → scikit-learn, MLflow, FastAPI
```

THE ONE RULE

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No topic is "done" until its mini project runs clean.
Speed means nothing. Depth means everything.
```

40+ mini builds. 6 checkpoint projects. 1 capstone. 180 days.